APOLLO HOSPITALS, GUWAHATI

HOSPITAL STAY AND DISCHARGE DELAY CLASSIFICATION:

A DATA ANALYSIS STUDY

Submitted By : ARNAB CHANDA

STREAM-HEALTHCARE ANALYTICS

UNIVERSITY-IIHMR UNIVERSITY

Under the Guidance of:

NAME-GOPAL UPADHYAY

DESIGNATION-BUSINESS ANALYST

Submitted to:

NAME-SUMAN MUKHERJEE

DESIGNATION-OPERATIONAL MANAGER

SIGNATURE-

ACKNOWLEDGMENT

I would like to express my gratitude to the management team at Apollo Hospitals for providing me with the opportunity to conduct my summer internship. I am incredibly thankful to the Operational Manager, Mr Suman Mukherjee and my hospital mentor Mr Gopal Upadhyay for their invaluable guidance, constant support, and mentorship throughout the duration of this project.

ABOUT THE ORGANIZATION

Apollo International Hospital, Guwahati is one of the leading tertiary care multi-specialty hospitals in Northeast India. The hospital was established in 2008 as International Hospital and later became Apollo Hospitals Guwahati through a joint venture between Apollo Hospitals Enterprise Limited (AHEL) and Assam Hospitals Ltd. (AHL), marking Apollo's first presence in Northeast India.

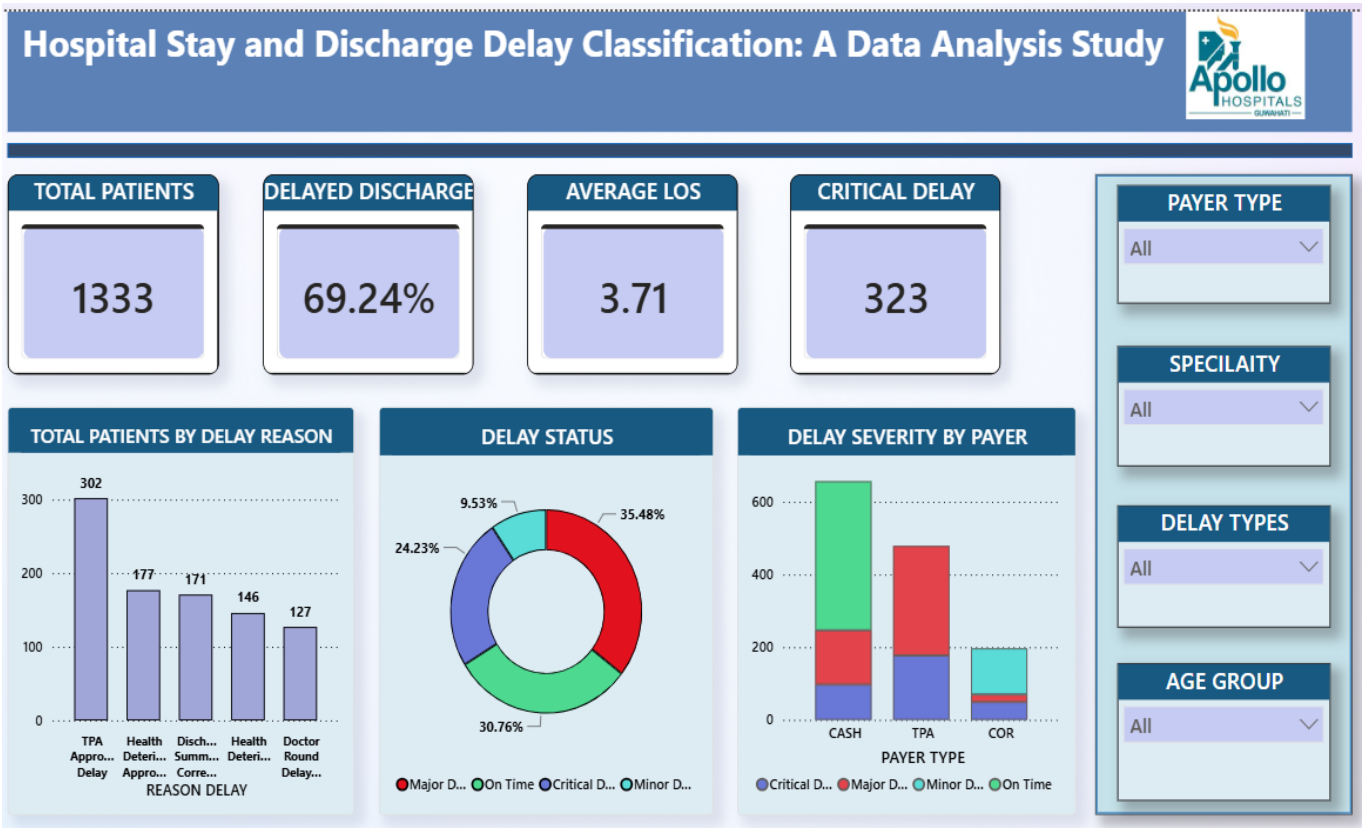
The hospital has a **230-bed** capacity and provides advanced healthcare services across a wide range of medical and surgical specialties.

The primary objective of the Hospital is to **deliver high-quality, patient-centric, accessible, and affordable healthcare through advanced medical technology, skilled healthcare professionals, and evidence-based clinical practices while** maintaining the highest standards of patient safety and clinical excellence**.**

PROJECT OBJECTIVE

- **Discharge Delay Analyse**- To find out the discharge delay across various specialities.
- **Payer Type and LOS** -To evaluate the relationship between payment methods and their corresponding impact on the average Length of Stay (LOS).
- **Departmental Bottlenecks**-To identify the specific medical specialities which are facing highest operational inefficiencies.

DISCHARGE ANALYTICS DASHBOARD



ABOUT THE DASHBOARD

- The Dashboard Hospital Stay and Discharge Delay Classifications represents 1 month discharge analysis across all specialities and the Length Of Stay(LOS) of patients in that 1 month period.
- The four boxes at the top provide an instant "pulse check" of the hospital's current efficiency.
- We can see the **Total Patient Load (1,333)** and the **Current Discharge Delay rate (69.24%)**, instantly. By highlighting **Average Length of Stay (3.71 days)** alongside **Critical Delays (323 cases)**, we can instantly spot if a specific department is becoming overcrowded or blocked.
- The charts move beyond just showing numbers, they reveal the underlying various causes of delays.

KEY INSIGHTS

- **The Core Operational Bottleneck: Third-Party Administrator (TPA) Approvals**
    1. Looking at the "Total Patients by Delay Reason" chart, **TPA Approval Delay** (302 cases) and **Health Deterioration/TPA Approval Delay** (177 cases) account for a massive **479 delayed patients**.
    2. This is huge bottleneck as large number of patients get recovered soon in the hospitals but they can't leave the high revenue generated bed due to waiting for insurance paperwork. Reducing this cycle directly lowers the overall **3.71 days Average LOS**.

- **Clinical Specialty Hotspots:**
    1. **Nephrology:** Highest average stay at **5.42 days** with a staggering **72.5% delay rate** followed by **General Medicine** with **69.9%** and **Gastroenterology** with **65.8%** delay rate.

- **The Financial Payer Disconnect-Cash vs. TPA**
    1. While **CASH** paying patients account for all **most "On Time" discharges** in the hospital, they also hold the longest ALOS at **3.98 days**. Conversely, TPA patients leave faster on average (**3.42 days**) but suffer from 100% administrative "Major" or "Critical" delays at the point of exit.
- **Internal Operational Red Flags-Doctors Rounds and Discharge summary Correction.**
    1. **Discharge Summary Correction**: Causes **171** delays (predominantly among CASH patients).
    2. **Doctor Round / Report Delays:** Accounts for **127 delays** (heavily concentrated among Corporate patients).

Doctor round delays mean discharges are being declared late in the afternoon or evening instead of during morning rounds. This pushes physical patient discharge past checkout times, freezing bed availability for incoming afternoon admissions.

PROBLEM IDENTIFIED

- **Administrative & Payer Bottlenecks: Insurance Approval Lag**
    1. TPA processing is the absolute largest single driver of discharge delays, impacting **479 patients.**
    2. The Final Billing Desk waits until the final discharge summary preparation, which causes the delay.
    3. Insurance providers operate on independent software systems from hospital management. This creates fragmented coordination between the hospital management and insurance providers and this leads to multi hours query cycles.

- **Operational Documentation Bottlenecks: Summary Corrections**
    1. **Discharge Summary Corrections** account for **171 major delay incidents**, creating operational efficiency down and driving up the average length of stay to **4.11 days** for affected individuals.
    2. Current documentation processing capacity is outpaced by the volume of daily patient discharges, creating an infrastructure-driven operational bottleneck.
    3. Fragmented coordination and the absence of real-time notification triggers between clinicians, ward secretaries, and data entry staff create data gaps. When a discharge summary is finalized, the lack of an automated alert system delays notification, coinciding with periods when clinicians are rightly occupied with high-acuity responsibilities in the OT or ICU.

- **Clinical Workflow Overlap: Delayed Doctor Rounds**
    1. **Doctor Round and Report Delays** cause **127 discharge failures**, directly leading to an ALOS of **3.56 days**.
    2. **Clinical vs. Administrative Misalignment:** Senior consultants naturally prioritize emergency admissions, active surgeries, and critical ICU rounds at the start of their shifts. Administrative tasks—such as signing physical discharge papers—are pushed to late afternoon or evening.

3. The Specialization Mix- High-volume, complex specialties like **General Medicine (71 Major Delays)** and **Nephrology (66 Major Delays)** run heavily interdependent care models. If a Nephrology patients wait for the cross-consultation report from the Cardiology department, the discharge summary cannot be finalised, creating multidepartment finalised.

Strategic Recommendation

- **Digitalize the Discharge Summary Workflow:** Upgrade from the manual, typewriter-reliant dependency to a templated, auto-populating module within the Electronic Health Record (EHR) system.

- **Early Draft Preparation**- Require ward secretaries and junior doctors to draft clinical discharge summaries after the patient OT by consulting with the primary consultant or during patient stay.

- **Establish a Dedicated TPA Desk at Wards:** Create a mini-TPA insurance desk on high-volume inpatient floors to verify documents 24–48 hours *before* clinical discharge, instead of starting the process at exit.

- **Cross consultation Service Agreement**: Enforce strict 2-hour response windows for cross-specialty references (e.g., Cardiology clearing a Nephrology patient) to prevent multi-department coordinate lags**.**

SWOT ANALYSIS

- **STRENGTH-**
    1. **Leading Regional Presence:** Positioned as a premier 214-bed tertiary care multi-specialty hospital with a strong brand presence in Northeast India.
    2. **Strong Clinical Infrastructure:** Equipped with advanced healthcare services across diverse medical and surgical specialties.

- **WEAKNESS-**
    1. **Severe Discharge Inefficiencies:** A high overall discharge delay rate of 69.24%, with 323 critical delay cases heavily impacting bed turnover.
    2. **Outdated Documentation Infrastructure:** Heavy reliance on manual typewriters to process daily discharges, creating a massive logistical slowdown at the ward desk.

- **OPPORTUNITIES-**
    1. **Standardized Cross-Consultation SLAs:** Implementing strict response-time windows for multi-specialty cross-consultations (e.g., Cardiology clearing a Nephrology summary) to prevent multi-day stays.
    2. **Optimized Payer Pathways:** Redesigning the insurance approval pipeline by introducing pre-authorization or early-morning documentation windows to clear TPA-related administrative lags.
- **THREATS-**
    1. **TPA Approval Delays**: Rigid external insurance approval lags that directly drive a massive chunk of operational delays, locking up 479 patients who are clinically ready to leave.
    2. **Bed Stagnation & Revenue Loss:** Extended Average Length of Stay (ALOS) caused by administrative delays forces recovered patients to occupy high-revenue beds, blocking incoming emergency admissions.

KEY LEARNINGS

- **Bed Turnover is Driven by Administration, Not Just clinical care:**
    1. A patient being "clinically cured" does not equate to an operational discharge. Administrative and insurance clearances heavily dictate hospital efficiency and turnover.

- **Identifying Bottlenecks via Data:**
    1. By analysing a dataset of 1,333 patients, I learned how to pinpoint exact operational bottleneck—discovering that a staggering 69.24% of patients experience discharge delays.

- **The Payer Mix Paradox:**
    1. I have analysed how different financial segments interact with hospital operations. While Cash patients exit smoothly but have a longer clinical ALOS (3.98 days), TPA (Insurance) patients have a shorter clinical stay (3.42 days) but suffer from 100% administrative "Major" or "Critical" delays at the point of exit.

- **Modern Infrastructural Changes:**
    1. I have learned how legacy processes can cripple modern efficiency, illustrated by the hospital's continued reliance on physical typewriters for discharge summaries, which directly caused 171 major delay incidents.

CONCLUSION

Discharge delays at Apollo Hospitals, Guwahati are driven by a combination of **external administrative lags, operational infrastructure, and unaligned clinical workflows**. While the hospital faces a **69.24% discharge delay rate** heavily anchored by external TPA insurance lags (479 patients) and outdated documentation tools, a significant bottleneck stems from clinical patterns—specifically, the practice of pushing discharge summaries to the late afternoon or evening due to active clinical priorities.